Machine Learning Project Report: Used Car Price Prediction

I. Introduction and Problem Definition

This project addresses the problem of predicting the selling price of used cars based on the provided `train.csv` and `test.csv` datasets, which are likely associated with the "Cars 4 You: We buy your car!" project. The objective is to build a simple, yet effective, machine learning model to estimate the car's price, which can be used by management to understand pricing dynamics and inform business strategy.

The core problem is a **Regression** task, as the goal is to predict a continuous numerical variable: the car's price. The combined dataset contains **108,540 records** with features such as `Brand`, `model`, `year`, `mileage`, `engineSize`, and `transmission`. The initial data exploration revealed the presence of **missing values** and **inconsistencies** (e.g., negative mileage, misspellings in categorical features), necessitating a robust preprocessing strategy.

II. Project Pipeline Schematic

The machine learning project followed a standard pipeline, broken down into five distinct stages:

Stage	Techniques Used	Purpose
1. Data Exploration	Descriptive Statistics, Missing Value Analysis, Visualization (Histograms, Box Plots)	To understand data structure, identify key relationships, and detect data quality issues (missing values, inconsistencies).
2. Preprocessing	Imputation (Median/Mode), Feature Engineering (`Car_Age`), Log Transformation, Categorical Clean-up, One-Hot Encoding, Standard Scaling	To clean the data, handle missing values and inconsistencies, create meaningful features, handle skewness/outliers, and prepare data for modeling.
3. Feature Selection	Recursive Feature Elimination (RFE)	To select a powerful subset of 20 features from the high-dimensional feature space (138 features) that maximizes model performance and interpretability.
4. Model Building	Train-Test Split, Multiple Linear Regression	To train a simple predictive model on the selected features.
5. Model Assessment	R-squared (R2), Root Mean Squared Error (RMSE)	To evaluate the model's predictive accuracy and generalization ability on unseen data.

III. Data Preprocessing and Feature Selection Details

A. Data Preprocessing (Task 2)

The preprocessing stage was critical for transforming the raw data into a format suitable for the Linear Regression model, with a focus on handling the newly identified data quality issues.

Variable	Technique	Justification
Missing Values	**Imputation (Median/Mode)**	Numerical features (`mpg` , `tax` , etc.) were imputed with the **median**, and categorical features (`Brand` , `model` , etc.) with the **mode** to fill in gaps and preserve dataset size.
price	**Logarithmic Transformation** (`np.log1p`)	Applied to the target variable to handle its right-skewed distribution and mitigate the influence of outliers.
year	**Feature Engineering** (`Car_Age`)	Converted to `Car_Age` (2025 - year). Car age is a more intuitive and direct measure of depreciation.
Inconsistencies	**Clean-up & Correction**	The 'transmission' typo 'anual' was corrected to 'Manual', and negative 'mileage' values were corrected using the absolute value.
Categorical Variables	**Grouping & One-Hot Encoding**	Low-frequency categories in `Brand` and `model` were grouped into 'Other' to reduce dimensionality. Remaining categorical features were converted using **One-Hot Encoding** with `drop_first=True` .
Numerical Features	**Standard Scaling**	Applied to numerical features to standardize them (mean=0, std=1), which is essential for the Linear Regression model.

B. Feature Selection (Task 3)

Given the large number of features (138) after one-hot encoding, **Recursive Feature Elimination (RFE)** with a **Linear Regression** estimator was used to select a powerful subset of **20 features**.

Justification for Final Selection: The RFE approach provides a data-driven method for selecting the features that are collectively the most predictive for the Linear Regression model. The selected features include key factors such as `engineSize` , `mpg` , `Car_Age` , and specific one-hot encoded categories for `model` and `transmission` , which are the primary drivers of car price.

IV. Model Building and Assessment (Task 4)

A **Multiple Linear Regression** model was chosen as the required "simple model" and trained on an 80/20 train-test split of the processed training data.

Assessment Metrics and Results:

Metric	Training Set	Test Set
R-squared (R2)	0.3511	**0.3377**
RMSE (Log Scale)	0.4295	0.4307
RMSE (Original Scale)	N/A	**8046.60**

The model achieved an **R-squared of 0.3377** on the test set, indicating that approximately 34% of the variance in the log-transformed price is explained by the model. The **Root Mean Squared Error (RMSE) on the original price scale is 8046.60**, which represents the average prediction error in the car's currency. The similar performance between the training and test sets confirms that the model is **well-generalized** and not overfitting.

The most significant features, as indicated by the model coefficients, are related to the **transmission type** (due to the presence of multiple, slightly different one-hot encoded columns for automatic/semi-automatic transmission) and key physical characteristics like `engineSize` and `mpg`.

End of Report
